Exp: 03

Study of Different Types of Network Cables and crimping of Cables with RJ45 Connector

Aim:-

To study different types of network cables and understand the crimping process using an RJ45 connector.

Concept:-

Network cables are used to connect computers and other networking devices for data communication. The most commonly used cable is CAT5 is the unshielded Twisted Pair (UTP) cables. An RJ45 Connector is attached to the ends of the cable of establish a network connection. The wires are arranged according to the T568A or T568B wiring standard before crimping and the completed cable is tested to ensure proper connectivity.

Types of Network cables :-

Cable Type	Description
1) UTP (Unshielded Twisted pair)	- Commonly used for Ethernet LAN's and office networks
2) STP (Shielded Twisted pair	- used in area with high electrical or electromagnetic interference
3) Coaxial cable	- used for cable television broad band internet, and CCTV systems
4) optical fiber cable	- used for high-speed high Bandwidth and long distance communication

Crimping procedure :-

1. Cut the CTP cable to the required length
2. Strip about 2-3cm of the outer insulation
3. Untwist and straight the wires.
4. Arrange the wires according to the T568B standard.

5) Trim the wire ends evenly

6) Insert the wires into the RJ45 connector.

7) Crimp the connector using a crimping tool.

8) Test the cable using a LAN cable tester.

observation :-

* The wires were arranged according to the T5 68B Standard.

* The RJ45 connector was crimped successfully

* The cable tester indicated proper continuity.

* The cable was ready for network communication.

Result :-

The different types of network cables were studied, and the UTP cable was successfully terminated using and RJ45 connector and tested for proper connectivity.

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